

Big Data Analytics Techniques

What is Big Data Analytics?

- Big data analytics refers to the complex process of extracting, transforming, exploring, and analyzing large data to uncover useful information that can be used to solve a certain business problem. It is the process used to extract meaningful insights, such as hidden patterns, unknown correlations, market trends, and customer preferences
- It is widely used across various industries such as health care, education, insurance, artificial intelligence, retail, and manufacturing to understand the useful information that can be used to improve business processes, systems functionalities, and market profitability.

Big data analytics can be used to:

- Understand customer behavior in order to optimize the customer experience
- Predict future trends in order to make better business decisions
- Improve marketing campaigns by understanding what works and what doesn't
- Increase operational efficiency by understanding where bottlenecks are and how to fix them

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Types of Big data analytics

Big data analytics can be classified into four types:

1. Descriptive analytics
2. Diagnostics analytics
3. Predictive analytics
4. Prescriptive analytics



DESCRIPTIVE

What happened?



DIAGNOSTIC

Why did it happen?



PREDICTIVE

What will happen next?



PRESCRIPTIVE

What's our course of action?

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Types of Big data analytics

Big data analytics can be classified into four types:

1. **Descriptive analytics:** This refers to data analytics that can be used to summarize data using numeric and graphic data summaries that can be easily read and interpreted. This data analytics helps to create reports and visualize information that can detail company profits and sales
2. **Diagnostics analytics:** This refers to data analytics that can be used to understand why a certain problem occurred. It the method which explains why and how something happened by identifying patterns and relationships in available data.
3. **Predictive analytics:** This refers to data analytics that looks at past and present data to make predictions. It the method which uses historical data to uncover patterns and make predictions on what's likely to happen in the future.
4. **Prescriptive analytics:** This refers to data analytics that provides solution to a problem by providing specific recommendations on what should be done better.

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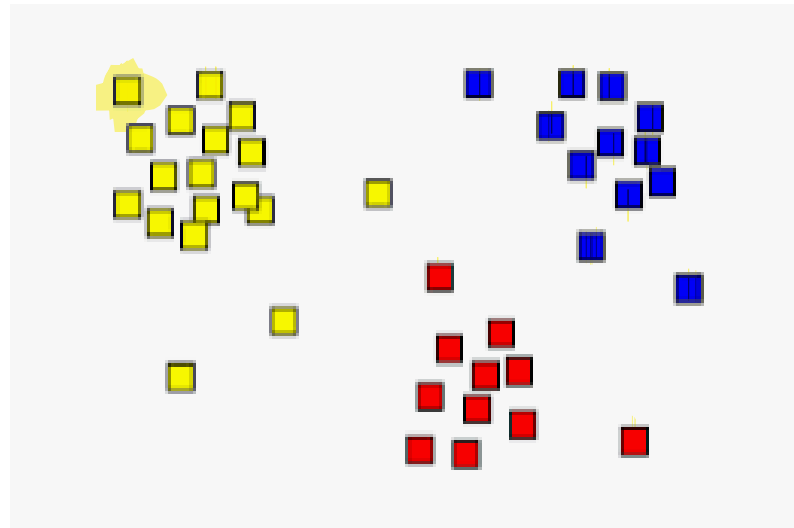
Some examples of Big Data Analytics techniques

Big data analytics techniques refers to the methods used to examine data and generate useful insights. The following are some examples of Big data analytics techniques:

Cluster analysis

This refers to the method which divides the input datasets into logical groups of related items. For example, grouping similar news articles, grouping similar customers based on their profile, etc.

It is the method which group the data points into different clusters, consisting of similar data points.



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Applications of Cluster Analysis

The clustering technique can be widely used in various tasks. Some most common uses of cluster analysis are:

Market Segmentation:

- This refers to process of dividing the target market into approachable groups.
- It is the way of creating subsets of a market based on demographics, needs, priorities, common interests, and other psychographic or behavioral criteria used to better understand the target audience

Anomaly detection:

- Anomaly detection (also known as outlier detection) is the identification of items, events or observations which do not conform to an expected pattern or behavior in comparison with other items in a given dataset

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Machine learning

- Machine learning is a method that enables a machine to automatically learn from data, improve performance from experiences, and predict things without being explicitly programmed.
- It is the method that allows systems to learn and improve automatically based upon experience.

Applications of Machine learning: The following are some applications of machine learning

Social Media Features

- Social media platforms use machine learning algorithms to create some attractive and excellent features.
- For instance, Facebook uses chats, users' activities, likes, and comments to make useful recommendations.

Product Recommendations

- Product recommendation is one of the most popular and known applications of machine learning.
- Machine learning uses customer behaviors based on the previous purchases, searching patterns, and cart history to make product recommendations.

Image Recognition

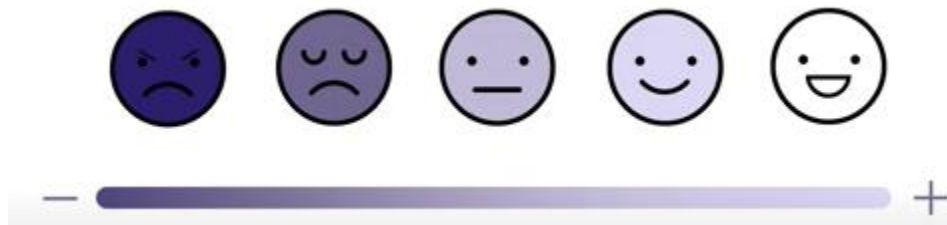
- Image recognition uses machine learning to detect features or an object in the digital image.

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Sentiment Analysis

is the process of analyzing digital text to determine if the emotional tone of the message is positive, negative, or neutral.

- **Sentiment analysis:** This refers to the process of computationally identifying and categorizing opinions from piece of texts and determine whether the write's attitudes toward particular topic or product, is positive, negative, or neutral.



- It is the method that helps to analyze and the understand the tone of text.
- It is the method of analyzing the emotions of a particular text, such as social media posts.
- It is the computational process which use Natural Language Processing and Machine learning to generate useful insights.



is a method in data analytics that divides a large dataset into smaller, more meaningful subgroups (segments) based on shared characteristics.

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Application of Sentiment Analysis

- Some popular sentiment analysis applications include social media monitoring, customer support management, and analyzing customer feedback.

Social media monitoring

- Social media posts often contain some of the most honest opinions about the products, services, and businesses

Customer support management

- Customer support management presents many challenges due to the sheer number of requests, varied topics, and diverse branches within a company

Brand monitoring and reputation management

- Bad reviews can increase online, and the longer you leave them the worse the situation will be.

User Comments



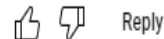
Moris Collins 35 minutes ago

Hawa ndio askari wetu wanavyopambana na uhatifu kazi nzuri askari wetu



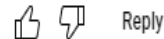
Mwaka 1 hour ago

Good Speech!!!



Chesco 1 hour ago

Good job



Md Ta 14 minutes ago

Safi sana



SanoureyAli WadoaKaroyo 59 minutes ago (edited)

Uyuu Kijana @ Godfrey Tomas Ngoba Unafanya kazi Zuli Pili Kwa Ustadi Wa Hali Ya juuu Sna Wewe Ni moja Kati Ya Mareporter Wazuli Sna Pili Unatoa Habari kwa Wakati Mnoo @Millard Ayo Muongezee Reporter wako Posho hakika Ni moja Ya Mareporter wako Wanao kuingizia Mapato kwa Wingi Mnoo

Milard Ayo
Channel